

B.Tech/M.Tech (Integrated) DEGREE EXAMINATION, JULY 2025

Fourth & Fifth Semester

21MAB302T - DISCRETE MATHEMATICS

(For the candidates admitted during the academic year 2021-2022 to 2024-2025)

Note:

- Part - A should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.
- Part - B and Part - C should be answered in answer booklet.

Time: 3 hours

Max. Marks: 75

PART - A (20 × 1 = 20 Marks)

Answer all Questions

Marks BL CO

- | | | | | |
|----|--|---|---|---|
| 1. | $A - B = B - A$ if and only if | 1 | 1 | 1 |
| | (A) $A \subset B$ | | | |
| | (B) $B \subset C$ | | | |
| | (C) $A \cup B = \phi$ | | | |
| | (D) $A = B$ | | | |
| 2. | Let $R = \{(6,2), (8,2), (10,2)\}$ be a relation from $B = \{6,8,10\}$ to $A = \{2,3,5\}$. Then R^{-1} is | 1 | 2 | 1 |
| | (A) $\{(2,6), (3,8), (5,10)\}$ | | | |
| | (B) $\{(2,8), (3,6), (5,10)\}$ | | | |
| | (C) $\{(10,2), (8,3), (6,5)\}$ | | | |
| | (D) $\{(2,6), (2,8), (2,10)\}$ | | | |
| 3. | Warshall's algorithm is applied to construct | 1 | 2 | 1 |
| | (A) Symmetric closure | | | |
| | (B) Transitive closure | | | |
| | (C) Reflexive closure | | | |
| | (D) Anti-symmetric closure | | | |
| 4. | If $f: R \rightarrow R$ given by $f(x) = 3x - 7$ then f^{-1} is | 1 | 3 | 1 |
| | (A) $7y - 3$ | | | |
| | (B) $3y + 7$ | | | |
| | (C) $\frac{y+7}{3}$ | | | |
| | (D) $\frac{y-7}{3}$ | | | |
| 5. | From a club consisting of 6 men and 7 women, in how many ways can we select a committee of 3 men and 4 women | 1 | 2 | 2 |
| | (A) 750 | | | |
| | (B) 700 | | | |
| | (C) 850 | | | |
| | (D) 600 | | | |
| 6. | If n pigeonholes are occupied by $kn + 1$ pigeons, where k is a positive integer, then atleast one pigeonhole is occupied by | 1 | 2 | 2 |
| | (A) k pigeons | | | |
| | (B) $k+1$ pigeons | | | |
| | (C) $k-2$ pigeons | | | |
| | (D) $k-3$ pigeons | | | |
| 7. | Among 30 Computer Science students, 15 know JAVA, 12 know C++ and 5 know both. How many students know exactly one of the languages | 1 | 3 | 2 |
| | (A) 27 | | | |
| | (B) 22 | | | |
| | (C) 17 | | | |
| | (D) 5 | | | |
| 8. | If LCM of two number is 14 and GCD is 1 then the product of two numbers is? | 1 | 1 | 2 |
| | (A) 14 | | | |
| | (B) 15 | | | |
| | (C) 7 | | | |
| | (D) 49 | | | |
| 9. | The statement $P \vee \neg P$ is a | 1 | 2 | 3 |
| | (A) contradiction | | | |
| | (B) tautology | | | |
| | (C) contrapositive | | | |
| | (D) inverse | | | |

- | | | | | | |
|---|---|------------------|---|---|---|
| 10. The dual of $\neg P \rightarrow (P \rightarrow Q)$ is
(A) $P \vee (\neg P \wedge Q)$
(C) $P \rightarrow \neg (P \rightarrow Q)$ | (B) $\neg (\neg P) \wedge (\neg P \wedge Q)$
(D) $(\neg P \wedge Q) \wedge \neg P$ | 1 | 2 | 3 | |
| 11. Contrapositive of $p \rightarrow q$ is
(A) $q \rightarrow p$
(C) $\neg q \rightarrow \neg p$ | (B) $\neg q \rightarrow p$
(D) $q \rightarrow \neg p$ | 1 | 4 | 3 | |
| 12. For each $n \in \mathbb{N}$, $3^{2n} - 1$ is divisible by | (A) 8
(C) 32 | (B) 16
(D) 18 | 1 | 2 | 3 |
| 13. The identity element In the group $G = \{2, 4, 6, 8\}$ under multiplication modulo 10 is
(A) 5
(C) 6 | (B) 9
(D) 12 | 1 | 2 | 4 | |
| 14. Let G be a group. If $a, b \in G$ then inverse of $(a * b)$ is
(A) $a^{-1} * b^{-1}$
(C) $a^{-1} * b$ | (B) $a * b^{-1}$
(D) $b^{-1} * a^{-1}$ | 1 | 4 | 4 | |
| 15. Every sub-group of a cyclic group is
(A) Homomorphic
(C) Isomorphic | (B) Cyclic
(D) Abelian | 1 | 1 | 4 | |
| 16. The Hamming distance between the codes $x = 010000$ and $y = 000101$ is
(A) 3
(C) 6 | (B) 2
(D) 5 | 1 | 5 | 4 | |
| 17. The maximum number of edges in a simple graph with n vertices is
(A) $\frac{n(n-1)}{2}$
(C) $\frac{(n+1)(n-1)}{2}$ | (B) $\frac{n(n+1)}{2}$
(D) $\frac{n}{2}$ | 1 | 2 | 5 | |
| 18. A connected graph is a Euler graph if and only if each of its vertices is of
(A) Odd degree
(C) Equal degree | (B) Even degree
(D) Increasing degree | 1 | 2 | 5 | |
| 19. Any connected graph with n vertices and $n-1$ edges is
(A) Graph
(C) Tree | (B) Closed graph
(D) Spanning tree | 1 | 4 | 5 | |
| 20. The chromatic number of the null graph is
(A) 4
(C) 3 | (B) 2
(D) 1 | 1 | 5 | 5 | |

PART - B ($5 \times 8 = 40$ Marks)

Answer **all** Questions

Marks BL CO

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|--|---|---|---|
| 21. (a) (i) Define Partially ordered set. (2 Marks)
(ii) Draw the Hasse diagram for the poset $\{D_{30}, \leq\}$ where $\leq$ is the relation is a divisor of 2. (6 Marks) | 8 | 2 | 1 |
|--|---|---|---|

(OR)

- (b) If $f: \mathbb{Z} \rightarrow \mathbb{N}$ is defined by $f(x) = \begin{cases} 2x - 1, & \text{if } x > 0 \\ -2x, & \text{if } x \leq 0 \end{cases}$

Prove that i) f is one-one ii) f is onto iii) Find f^{-1}

22. (a) Use Euclidean algorithm find the integers x and y such that $154x + 250y = 2$ 8 4 2

(OR)

- (b) Find the number of integers between 1 and 250 both inclusive that are not divisible by any of the integers 2, 3, 5 and 7.
23. (a) Prove that the following set of premises is inconsistent. 8 3 3
- (i) If Rama gets his degree, he will go for a job.
 - (ii) If he goes for a job, he will get married soon.
 - (iii) If he goes for higher study, he will not get married.
 - (iv) Rama gets his degree and goes for higher study.

(OR)

- (b) Use mathematical induction to prove that $8^n - 3^n$ is divisible by 5, for $n \geq 1$.
24. (a) Show that $(Z, +, \cdot)$ is a commutative ring with identity where the operations $+$ and $\cdot$ are defined for any $a, b \in Z$ as $a + b = a + b - 1$ and $a \cdot b = a + b - ab$ 8 5 4

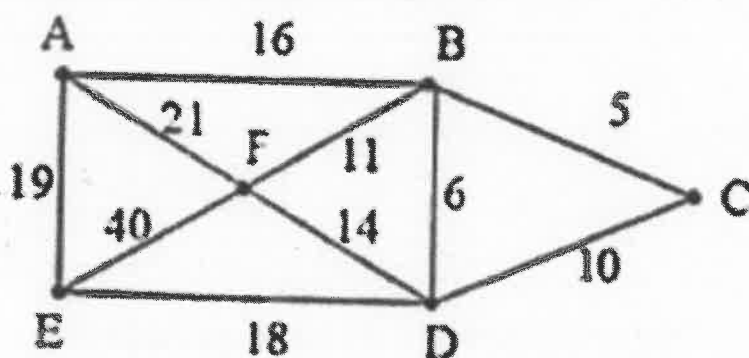
(OR)

- (b) Let $H = \begin{pmatrix} 0 & 1 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ be a parity check matrix. Determine the $(2,5)$ group code function $e_H: B^2 \rightarrow B^5$.

25. (a) Prove that the maximum number of edges in a simple disconnected graph G with n vertices and k components is $\frac{(n-k)(n-k+1)}{2}$ 8 3 5

(OR)

- (b) Find the minimum spanning tree for the weighted graph using Kruskal's algorithm. Also determine the minimum total weight.



PART - C (1 × 15 = 15 Marks)

Answer any 1 Questions

Marks BL CO

26. Given the generator matrix $G = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{pmatrix}$ corresponding to the encoding 15 3 4

function $e_G: B^4 \rightarrow B^7$. find the corresponding parity check matrix and use it to decode the following received words and hence to find the original message 110 0001, 1110111, 0010001, 0011100.

27. (i) If A, B and C are sets, prove analytically, that 15 3 1
 $A - (B \cap C) = (A - B) \cup (A - C)$. (5 Marks)

(ii) Find the transitive closure of the relation $R = \{ (1,1), (1,2), (2,1), (2,2), (3,3), (3,5), (4,3), (4,4), (4,5), (5,4), (5,5) \}$ defined on the set $A = \{1,2,3,4,5\}$ using Warshall's algorithm. (10 Marks)

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